Campus Item Exchange

Phases 4 and 5 Architecture and Detailed Design | 30 October 2026 | Academic prototype

Purpose and scope

An internal campus system for finding, requesting, transferring, maintaining and tracking reusable assets. The prototype covers one campus, two sample departments and about 20 labelled items. AI supports people with advice; permissions, stock, approvals and calculations remain controlled by application rules.

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Review and individual contributions

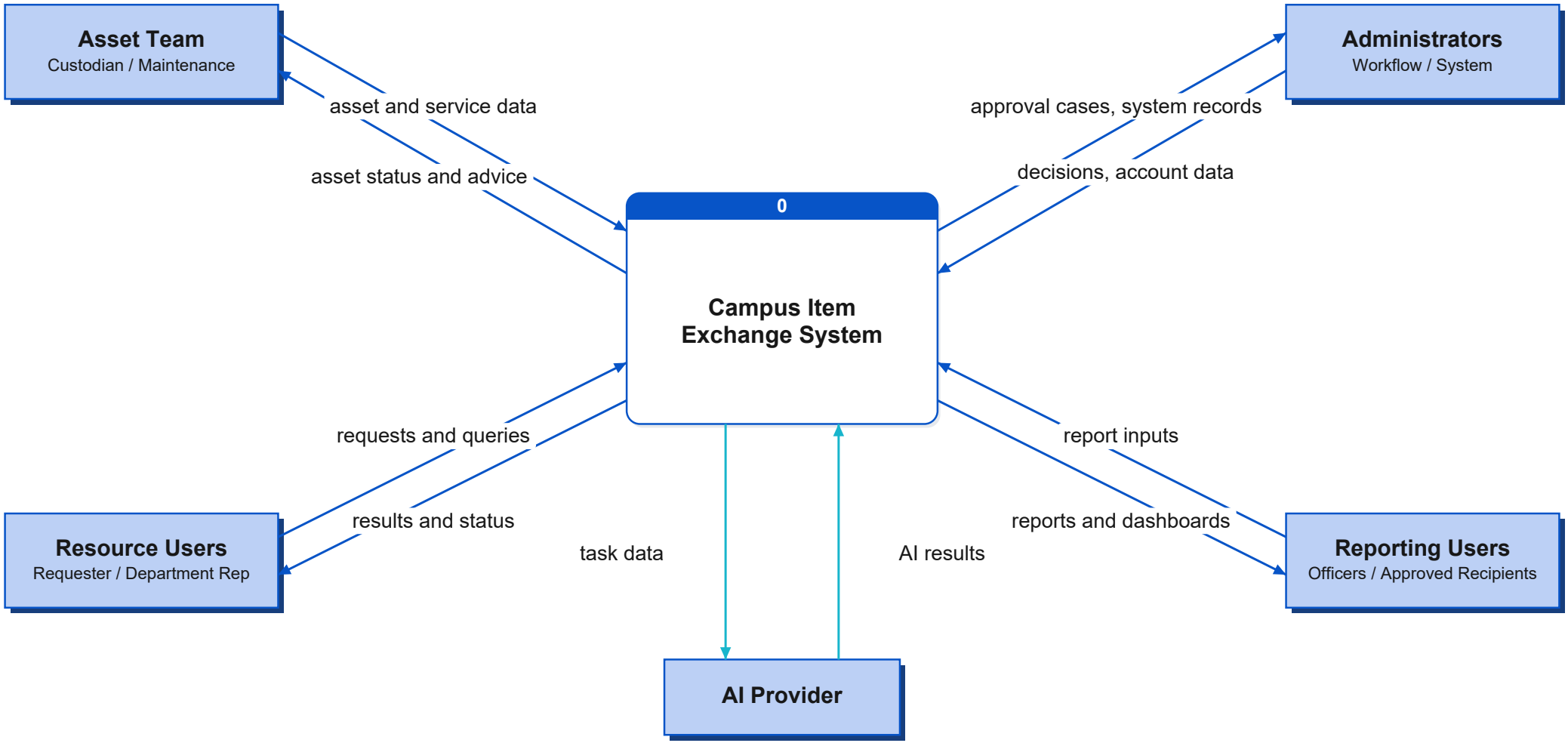
The four members and IDs match Phases 1-3. Each student must review and revise their contribution and commit it using their own identity. This document does not assert that individual GitHub commits or implementation tests have been completed.

Requirements and document status

Phase 3 requirement IDs appear in every section. Example records, AI outputs and screen values are synthetic. Deadlines, review gates and acceptance checks are proposed design settings; measured implementation results belong in later test records. This covers Phases 4 and 5.

Baseline: Campus Item Exchange Phase Three Requirements Document, 4 October 2026; Phase 4 Architecture Design lecture, Fall 2026.

Campus Item Exchange context diagram

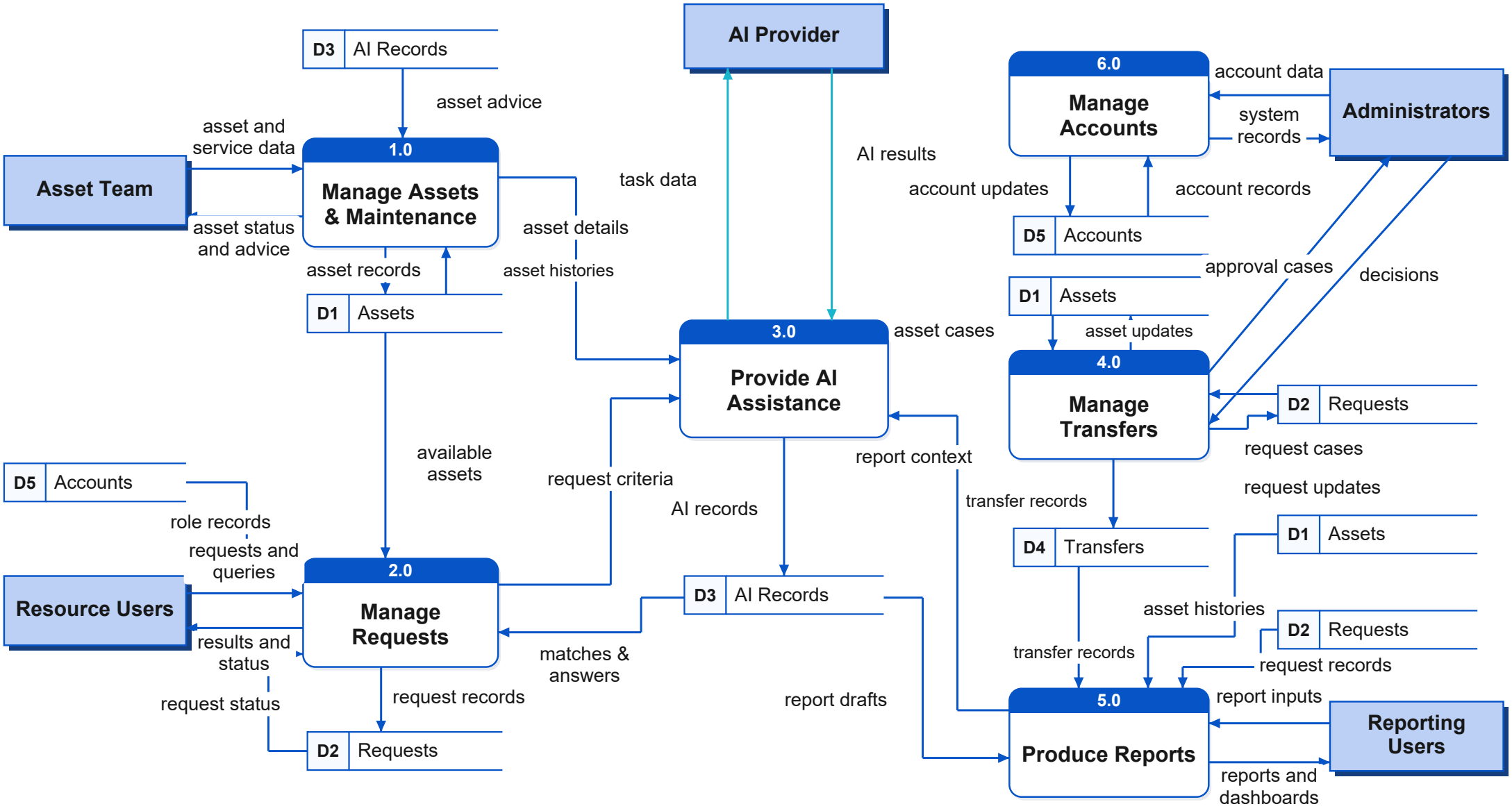


One process (0), grouped noun flows, no internal steps or data stores.

Reporting officers are Procurement, Finance and Sustainability. Grouping roles does not combine their permissions.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

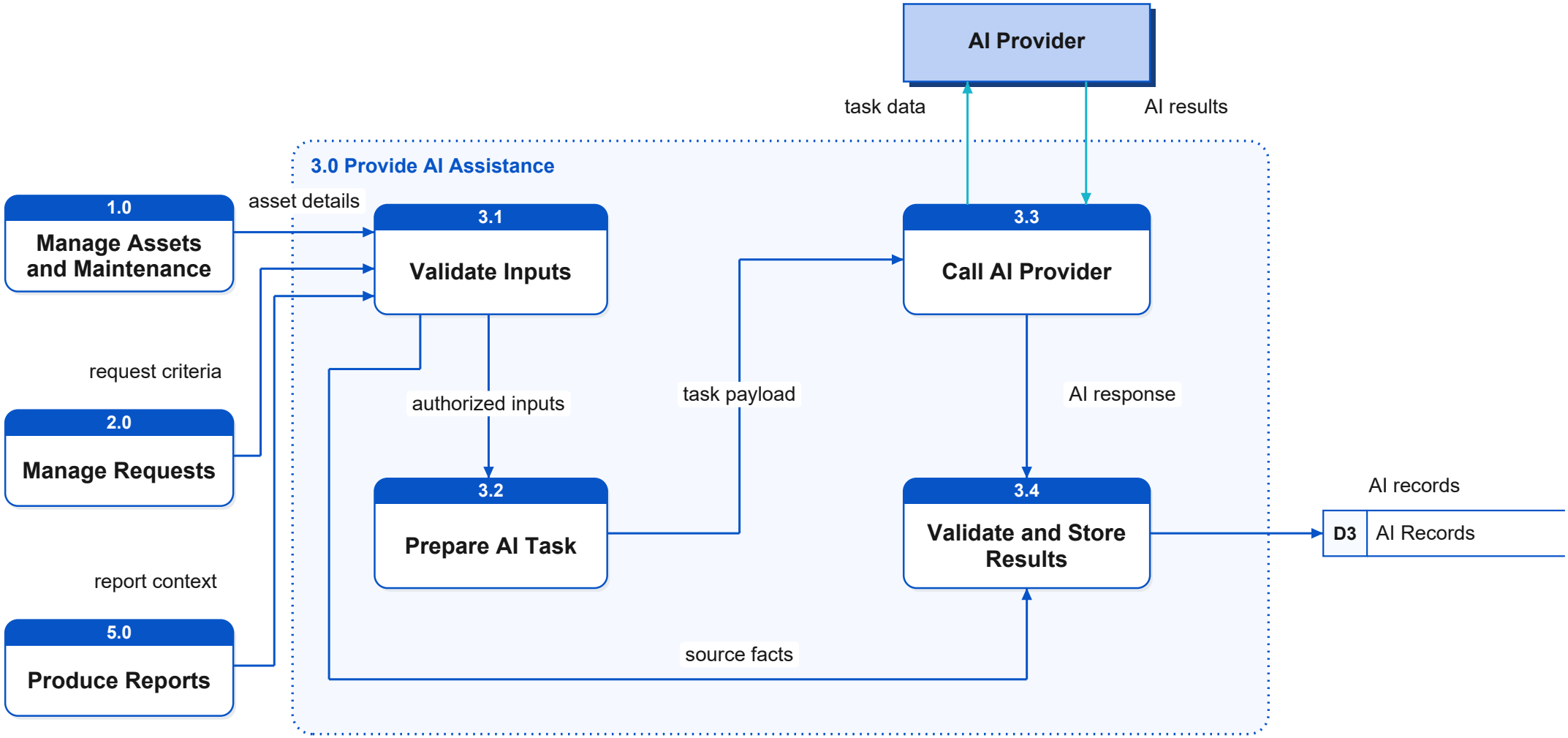
Campus Item Exchange Level 1 DFD



Repeated D-numbers refer to the same store. Maintenance is included in 1.0; approvals and disposition are included in 4.0.
Role families cover the full project. Sign-in, audit and notifications remain shared controls; permissions stay separate.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 2 of 3.0



The six flows crossing the dotted boundary are exactly the six flows of 3.0 on Level 1.

3.0 is expanded because it handles untrusted AI responses. Validated results or failure status are stored before returning to callers.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 1 balancing table

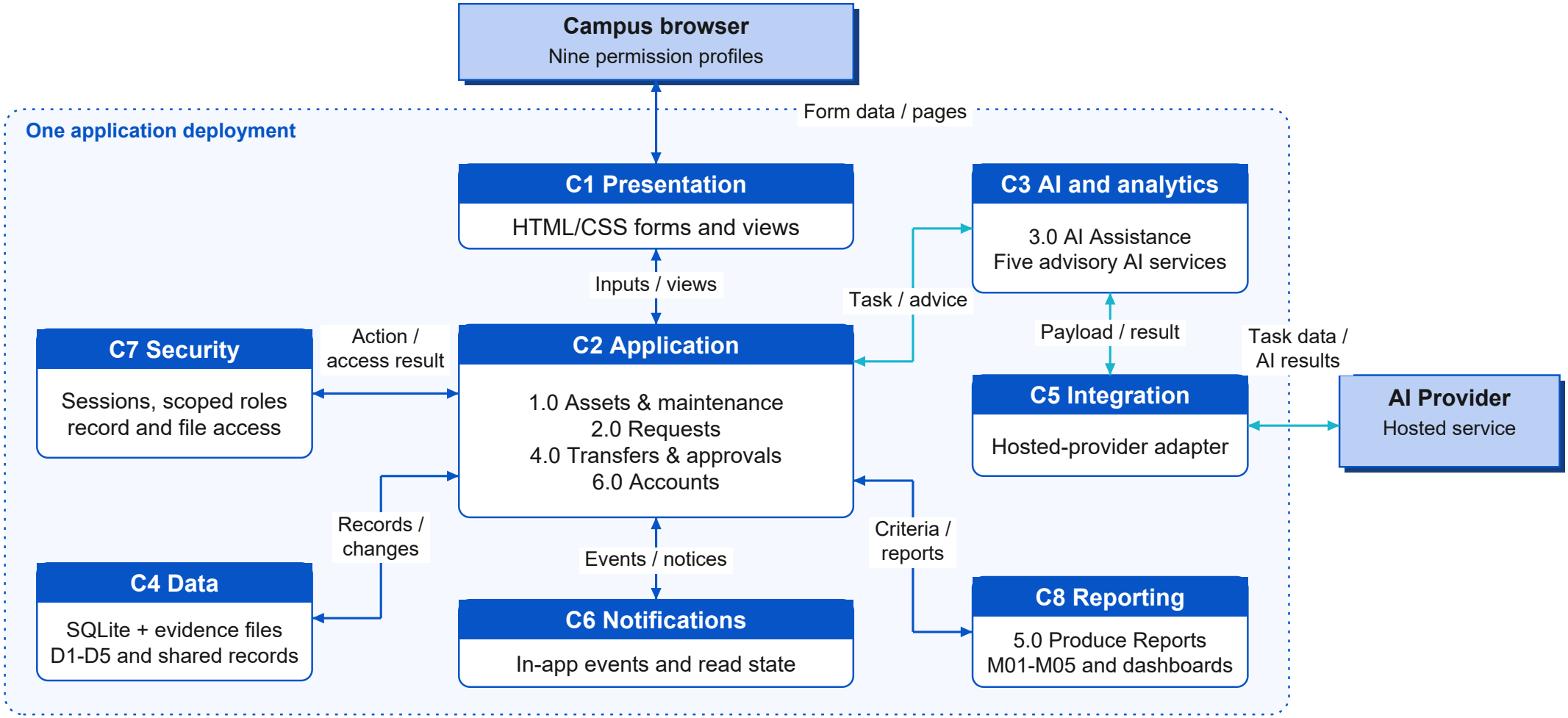
#	Context flow	External entity	In/out	Level 1 process	OK
1	asset and service data	Asset Team	in	1.0	✓
2	asset status and advice	Asset Team	out	1.0	✓
3	requests and queries	Resource Users	in	2.0	✓
4	results and status	Resource Users	out	2.0	✓
5	decisions, account data	Administrators	in	4.0, 6.0 (split)	✓
6	approval cases, system records	Administrators	out	4.0, 6.0 (split)	✓
7	report inputs	Reporting Users	in	5.0	✓
8	reports and dashboards	Reporting Users	out	5.0	✓
9	task data	AI Provider	out	3.0	✓
10	AI results	AI Provider	in	3.0	✓

VERIFY 10 of 10 match: same name, same direction, same entity. Grouped flows (5, 6) split at Level 1.

In/out is relative to the whole system. Flows 5 and 6 split between 4.0 Manage Transfers and 6.0 Manage Accounts.

3.1 Campus Item Exchange architecture view

Modular monolith: one Flask application, a private data environment and one external AI provider.



Routes call services; services use scoped repositories and coordinate AI and reports. Security applies to every server component.

Manual search, requests, asset history and numeric reports remain available during an AI failure. Paired labels show exchanged data.

Drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR09-NFR11; BR01-BR07.

3.2 Architecture decision records

Each record compares two options, explains the choice and trade-off, and states a planned verification check.

ADR-01 Architecture style

Drivers: FR01-FR18; NFR05, NFR12; BR01-BR07.

Option A
Layered monolith: one application organized by presentation, business logic and persistence.

Option B
Modular monolith: one application organized around business responsibilities, with shared services.

Choice We choose a modular monolith because six DFD responsibilities give the team clear module boundaries while reservations and transfers can share local transactions.

Give up We give up the simpler layer-only organization. Module boundaries need discipline; deployment, restarts and scaling remain shared.

Check Trace 1.0-6.0 to their owning components. Check module interfaces and test reservation and transfer rollback without partial changes.

ADR-02 Presentation and application stack

Drivers: Phase 3 Section 1 baseline; FR11, FR16; NFR02-NFR04, NFR07-NFR08, NFR12.

Option A
Flask with server-rendered HTML/CSS, ordinary forms and limited browser scripting where needed.

Option B
Flask JSON API plus a separate React browser application, with coordinated client and API behavior.

Choice We choose Flask with HTML/CSS because it preserves the Phase 3 baseline and keeps the four-person prototype manageable. All forms call the same authorized application services.

Give up We give up a separately maintained client application and some browser interaction flexibility; additional clients would need API endpoints.

Check Test login, request and approval flows in the recorded Chrome and Firefox versions at 1280 x 720, including keyboard use and forbidden direct requests.

3.3 Data and AI decision records

The technology choices preserve the Phase 3 prototype baseline. Provider capabilities still require confirmation.

ADR-03 Prototype data and evidence storage

Drivers: Phase 3 Section 1 baseline; FR05, FR08, FR12-FR13; NFR05, NFR09, NFR15.

Option A
SQLite and protected evidence files, with scoped repositories, explicit transactions and coordinated snapshots.

Option B
PostgreSQL and protected evidence storage, with a separate database service to configure and recover.

Choice We choose SQLite because the baseline is one campus, two sample departments and about 20 items. Transactions must recheck availability and prevent duplicate reservations or handovers.

Give up We give up an independently deployed database service. The prototype must pass the required concurrent-write and growth checks before acceptance.

Check Race requests for the last unit; retry a handover; restore database and evidence within 10 minutes. Run NFR15 with 200 items and 10 sessions.

ADR-04 AI gateway and provider integration

Drivers: AI01-AI08; NFR03-NFR04, NFR06, NFR10, NFR14; FR05, FR17.

Option A
One internal AI gateway and provider adapter: classification, semantic matching, ranking, action advice and LLM assistance/report drafts.

Option B
Direct provider calls in each feature module, with separate validation, privacy, timeout and fallback handling.

Choice We choose one gateway because shared task contracts keep authorized payloads, output validation and provenance consistent. AI suggests; application rules and human approvals control changes.

Give up We give up direct feature-specific provider access and accept a common dependency for all AI features. Contract changes require coordination.

Check Test all five services with success, timeout and invalid output. Enforce 15 seconds per AI call; preserve manual workflows and model/review records.

AI-use statement: AI assisted with drafting, figures and consistency checks. Team review and implementation testing are pending.

4.1 AI service plan

A = ask and wait. All services run through 3.0 Provide AI Assistance and the shared provider adapter.

Service	Pattern	Why	Time limit	If down or invalid	If unsure
Asset classification AI02	A	Custodian is editing one asset and can continue manually.	8 s	Preserve draft; select category and tags manually. Explain unsupported evidence.	Below 0.80 or confidence Unavailable: manual-review warning. Never pre-accept; unknown facts stay Unknown.
Semantic matching AI01	A	Small dataset; compare current authorized candidates without a maintained vector index.	At most 6 s; within shared 12 s	Keep need and filters; show labelled keyword/rule results. Skip AI ranking after matching failure.	After hard checks, compatibility below 75/100 means weak match. Unavailable confidence needs review; no default choice.
Ranking and explanations AI06	A	User needs one ordered list from the same eligible candidate set.	At most min(6 s, shared time left)	Keep candidates in labelled rule order; show known factors without invented AI scores.	No top-pick emphasis if top score <75, top-two gap <10, required fact missing or confidence <0.80/Unavailable.
Sustainable-action advice AI03	A	User asks about one asset; its ordinary service/disposition form remains available.	12 s	Preserve evidence and use normal maintenance/disposition forms; old advice shows its date and stale warning.	Below 0.80, Unavailable confidence or missing evidence: qualify advice. No invented saving or safety clearance; human approval stays required.
LLM assistant and report drafts AI04-AI05	A	Person requests an answer or optional draft; numeric reports appear independently first.	12 s per explicit answer or separate draft	Retain question; offer normal screens. Keep numeric report visible and label the narrative unavailable.	Cite sources or say unavailable. Reject false IDs, policy or totals. <0.80/Unavailable requires review; requests remain previews until confirmed.

Budgets include retrieval, provider transport and validation. The Phase 3 limit is 15 seconds per AI call.

Drivers: AI01-AI08; FR01-FR05, FR09-FR10, FR13, FR18; NFR03, NFR06, NFR10, NFR13-NFR14; UC09-UC13.

4.2 Shared AI controls and verification

Thresholds are proposed prototype review gates. Confidence and compatibility have different meanings.

Deadline

Every budget includes scoped retrieval, provider transport and validation. Matching and ranking share one 12-second deadline: matching <=6 seconds, ranking <=min(6, remaining). No fresh deadline starts between stages. NFR06 remains 15 seconds per AI call.

Failure and capacity

No automatic retry or unbounded waiting. Return AI busy promptly if no configured capacity is available. Reserve web capacity for ordinary screens. A new attempt needs an explicit action; late results cannot replace a timeout or change records.

Confidence

Use a numeric confidence only if its origin and 0-1 scale are documented for that task. The proposed warning threshold is 0.80. Otherwise show Unavailable and require manual review; never invent confidence or treat an LLM's self-reported number as calibrated accuracy.

Compatibility

The proposed normal-match threshold is 75/100. Top-pick emphasis also needs a 10-point gap when two candidates exist, complete required compatibility facts and no unresolved confidence warning. These are display/review gates, not accuracy claims or permissions.

Validation and authority

Check current identity, record scope, source revision, response schema, IDs, categories, facts and report totals. Recheck stock and safety when a user confirms. No score can override a failed rule, authorize a disposition or clear an unsafe asset.

Provenance

Store input references/snapshots, time, provider, model/version or Version unavailable, prompt version, output, confidence or Unavailable, validation outcome, and accepted/edited/rejected/overridden decision with reason. Exclude passwords, session secrets and AI keys.

Planned acceptance checks

Demonstrate successful results for all five services, not only fallback screens. Test at least 8/10 suitable top-three matches and 8/10 correct labelled classifications, including supported image/PDF input.
Check ten explanations against real factors; test department-name swaps, fabricated IDs, wrong totals, revoked roles, stale inputs and late/duplicate responses. Record tester, environment, date, expected/actual outcome and evidence.
Before acceptance, confirm provider/model versions, evidence handling, documented confidence semantics and actual capacity. Numeric reports remain available before a separately requested narrative draft.

AI07-AI08; FR12, FR16-FR17; NFR03-NFR06, NFR10, NFR13-NFR14; BR01, BR04-BR07

4.3 AI pattern decision records

A = ask and wait; B = compute ahead; C = queue and notify. Each service has a real alternative and a planned check.

Record / service	Two options	Choice and because	What we give up	Planned check
ADR-05 Classification	A: interactive call; B: recompute after each asset revision.	We choose A because the custodian reviews fields while editing.	We give up ready-made suggestions and completion after leaving the editor.	Test valid text/image/PDF, unsupported evidence, timeout and retained draft.
ADR-06 Semantic matching	A: compare on demand; B: maintain versioned representations on save.	We choose A because about 20 items do not yet justify index and invalidation handling.	We give up instant reuse of precomputed vectors; revisit if growth/latency checks fail.	Use ten known needs; test permission changes, stale stock and matching timeout.
ADR-07 Ranking	A: rank the current list; C: queue ranking and notify later.	We choose A because the requester is waiting to choose from one current eligible list.	We give up deferred results after the user leaves and accept a bounded wait.	Test top-two ties, unknown factors, department swaps and the shared deadline.
ADR-08 Sustainable-action advice	A: explicit foreground advice; C: queue and notify when ready.	We choose A because one-asset reviews fit the prototype and manual forms provide continuity.	We give up advice that completes after navigation away and accept a short wait.	Test missing evidence, timeout, stale advice and unchanged approval/safety controls.
ADR-09 Assistant and report drafts	A: separate requested answer/draft; C: queue longer generation and notify.	We choose A because it keeps one interaction path while numeric reports remain immediately usable.	We give up deferred completion and hidden long-running generation.	Test cited answers, exact request preview, mismatched totals and draft failure after report display.

Drivers: AI01-AI08; NFR01, NFR03, NFR05-NFR06, NFR10-NFR11, NFR14-NFR15; UC09-UC13. All settings require validation.

5.1 Human review and approval points

AI may suggest, rank, explain or draft. A permitted person decides; server checks remain mandatory.

Point	Who decides	What they review and confirm	Why / required record
H1 Metadata	Assigned asset custodian	Review entered fields, evidence, suggested category/tags, confidence and missing facts. Accept, edit or reject; confirm final fields before saving.	Risk: invented or incorrect metadata. Record input revision, reviewed values, AI provenance and edit/override reason. FR01-FR02; AI02, AI07-AI08.
H2 Request or chat preview	Requester / department representative, within scope	Review exact asset, quantity, destination, date, need and stock. Explicitly confirm: normal rules create an atomic hold. Choosing a match alone reserves nothing; unmatched needs may stay Seeking.	Risk: stale or unauthorized allocation. Record exact preview, creator, stock check, resulting request/hold or failure and AI review. FR03-FR05; AI01, AI04, AI06-AI08; BR01-BR02.
H3 Publication	Workflow administrator; no self-approval	Review release proposal, owner, condition, serviceability and evidence. Approve or reject with a reason; only approved, serviceable active units may be published.	Risk: publishing unsafe or unreleased stock. Record actor, time, reason and old/new publication state. FR02, FR17; BR01, BR04; NFR13.
H4 Reservation / transfer decision	Workflow administrator; no self-approval	Review purpose, destination, quantity, hold/expiry and current eligibility. Approval retains the same hold and changes no ownership. Rejection releases stock once.	Risk: unsupported authority or double deduction. Record case/hold IDs, actor, time, reason and status transition. FR05, FR07, FR17; BR02-BR04; NFR05.

Every gate records actor, time, case ID, reason and the previous/new state. No one approves their own request.

Drivers: FR01-FR08, FR12-FR13, FR16-FR17; AI01-AI02, AI04, AI06-AI08; BR01-BR04; NFR03, NFR05, NFR13.

5.2 Physical actions, reporting and accounts

Human judgment and evidence control changes that AI cannot authorize or verify.

Point	Who decides	What they review and confirm	Why / required record
H5 Collection and receipt	Source and receiving custodians	Source records collection arrangements and evidence. Receiver confirms the full approved receipt. Only then does one transaction update ownership/custody. Refused or overdue receipt enters dispute; units stay blocked.	Risk: unproven physical handover. Record both actors, times, evidence, quantity/condition and old/new custody, owner, location and hold. FR08, FR13, FR17; BR02-BR06.
H6 Repair / safety	Custodian authorizes repair; assigned maintenance staff records inspection	Review defect, job state, costs or Unknown and AI advice. Follow service states and record human inspection clearance before restoring serviceability. AI cannot clear an asset.	Risk: unsafe units returning to use. Record authorization, inspection evidence, service actions, condition and advice override reason. FR09, FR14; AI03; NFR13; UC07, UC11.
H7 Disposition	Custodian proposes/executes; workflow administrator approves	Review unallocated units, holds, condition, costs and evidence. Approve/reject with a reason. Retirement blocks use; final disposition requires execution evidence. No self-approval.	Risk: irreversible loss or disposal of held stock. Record proposal, review, approval, actors/times, execution evidence and lifecycle transition. FR10, FR13, FR17; BR04-BR07.
H8 Reporting	Procurement, finance or sustainability officer; workflow admin for operations	Validate only assigned inputs. Review period, record links, calculated totals, coverage and optional AI draft. Reject mismatched figures; numeric reports remain usable without AI.	Risk: invented impact or wrong period. Record input changes, contributing IDs, calculation basis and narrative review. No additional report-approval state is introduced. FR12, FR18; AI05; BR07; NFR11.
H9 Accounts / roles	System administrator	Review identity, active status, profile, department and custody. Confirm changes; reassign custody before deactivation. This profile alone cannot approve business cases.	Risk: excess access or orphaned custody. Record target account, actor, time, reason and old/new role, department and active state. Exclude secrets. FR11-FR12, FR14, FR16; BR06.

Workflow administrators resolve receipt disputes: acknowledged receipt completes; evidenced physical return permits cancellation and stock restoration.

Drivers: FR08-FR18; AI03, AI05, AI07-AI08; BR02-BR07; NFR03-NFR05, NFR11, NFR13; UC06-UC08, UC11, UC13.

5.3 Human review design decision

The interaction changes with the evidence and authority needed at each gate.

Record ADR-10	Stage-specific human review
Drivers	FR07-FR08, FR10, FR17; AI07-AI08; BR04-BR06; NFR13.
Option A	One generic human decision form for all approval gates, with case-specific validation.
Option B	Stage-specific review forms for metadata, requests, approvals, receipt and disposition, using shared authorization and audit services.
Choice and because	We choose stage-specific review forms because each gate needs different evidence, role checks and consequences. Explicit confirmation invokes the ordinary transaction; AI never applies an action.
What we give up	We give up one reusable generic screen and accept more form states and validation paths.
Planned check	Try a self-approval, an AI-only submission, a receipt without evidence, an unsafe unit and a duplicate confirmation; verify the required gates and atomic effects.

Failure and concurrency behavior

If permission, state, safety, stock or evidence changes before confirmation, reject the action and preserve valid input. Failed writes leave no partial update; repeated confirmation returns the existing result. A record correction appends a linked entry rather than erasing history.

Drivers: FR05-FR08, FR10, FR12-FR13, FR16-FR17; AI07-AI08; BR04-BR06; NFR05, NFR13.

6.1 Access matrix - requests and assets

Default: deny. A scope entry still requires an active account, permitted record, valid action and allowed state.

Action	RQ	DR	AC	MS	WA	PO	FO	SO	SA	AI
Sign in; read own notifications	Self	Self	Self	Self	Self	Self	Self	Self	Self	-
Search permitted catalogue listings	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Scope	Suggest
Create/view/cancel needs and requests	Own	Dept	-	-	-	-	-	-	-	Suggest
Register/edit assets; confirm category	-	-	Asset	-	-	-	-	-	-	Suggest
Propose release or disposition; record execution	-	-	Asset	-	-	-	-	-	-	Suggest
Decide approvals, disputes and Approved cancellation	-	-	-	-	Gate	-	-	-	-	-
Record collection; confirm receipt	-	-	Asset	-	-	-	-	-	-	-

RQ Requester | DR Dept. representative | AC Asset custodian | MS Maintenance staff | WA Workflow administrator
PO Procurement officer | FO Finance officer | SO Sustainability officer | SA System administrator | AI AI services

- Denied; Self own identity/recipient notices; Scope authorized listings; Own own request; Dept department requests (only creator cancels Seeking/Pending); Asset assigned/source/receiving assets; Gate eligible case with no self-approval.

AI describes suggestions, not a profile or database grant. It receives only the task data already authorized for the caller; it cannot execute a business write.

Drivers: FR01-FR18; AI01-AI08; BR03-BR06; NFR03-NFR04, NFR09-NFR10, NFR13; Phase 3 Section 5 permission profiles.

6.2 Access matrix - service, reporting and administration

Default: deny. A scope entry still requires an active account, permitted record, valid action and allowed state.

Action	RQ	DR	AC	MS	WA	PO	FO	SO	SA	AI
Record inspection/service/condition	-	-	Asset	Job	-	-	-	-	-	Suggest
Authorize repair	-	-	Asset	-	-	-	-	-	-	Suggest
View numeric reports; review AI draft	-	-	-	-	Ops	Demand	Finance	Env	-	Draft
Validate reporting inputs	-	-	-	-	-	Demand	Finance	Env	-	-
Manage accounts/profiles/departments	-	-	-	-	-	-	-	-	System	-
Read audit history	-	-	-	-	Ops	-	-	-	System	-
Backup and restore	-	-	-	-	-	-	-	-	System	-

RQ Requester | DR Dept. representative | AC Asset custodian | MS Maintenance staff | WA Workflow administrator
PO Procurement officer | FO Finance officer | SO Sustainability officer | SA System administrator | AI AI services

- Denied; Asset assigned assets; Job assigned service job; Ops operational reports/history; Demand demand and avoided-purchase fields; Finance financial fields; Env environmental/life-extension fields; System account, audit and recovery scope; Draft narrative only.

Custodians/requesters see permitted record history in normal screens. Audit-interface access is limited to reviewers. Evidence and notification links always recheck current parent-record permissions.

Drivers: FR01-FR18; AI01-AI08; BR03-BR06; NFR03-NFR04, NFR09-NFR10, NFR13; Phase 3 Section 5 permission profiles.

Campus Item Exchange | Phase 4 design | Synthetic examples; implementation checks are planned.

6.3 Security controls and enforcement

One central policy and scoped repositories protect every route, file, notification link and AI retrieval.

Control	Design and requirement references
One authorization policy	Use central action/role/scope checks and scoped repositories for protected pages, writes, evidence, notification links and AI retrieval. Recheck active state and current authority on every protected action. FR11, FR13, FR16; BR06; NFR03
Credentials and sessions	Store salted adaptive password hashes. Use HttpOnly session cookies, SameSite=Lax, Secure beyond localhost and CSRF tokens on state-changing forms. Keep AI keys and session secrets out of tracked files, reports, logs and AI payloads. Browser traffic beyond localhost and hosted AI calls use authenticated HTTPS where applicable. NFR04; Phase 3 Section 5 interfaces
Evidence validation	Accept JPEG, PNG and PDF up to 5 MB; reject unauthorized, unsupported or oversized files. Use randomized filenames outside the public web directory; retrieve through an authorized parent-record route; record uploader/time. FR13; NFR03-NFR04
AI isolation	Send only authorized task fields; treat descriptions/documents and returned text as data. Validate IDs, categories, facts, policy sources and report totals. AI has no approval, custody, account or disposal authority. AI07-AI08; NFR03, NFR10; BR04-BR07
Record integrity	Use parameterized SQL. Reservations, handover and disposition use atomic transactions, current-state checks and duplicate-action protection. Append corrections rather than deleting history. Use labelled synthetic data and retain course audit records. FR05, FR08, FR12; BR06; NFR05, NFR10

ADR-11 Authorization enforcement

Options: repeated route-local checks OR central policy plus scoped repositories.
We choose central checks because all nine profiles and AI retrieval must obey the same scope and active-account rules. We give up feature-local policy freedom and accept a shared dependency.
Planned check: test permitted actions and forbidden direct URLs/writes, cross-department data, evidence and AI retrieval for every profile; deactivate an account and retry.

Drivers: FR11-FR13, FR16; AI07-AI08; BR06; NFR03-NFR05, NFR10. Database and evidence remain private; AI has no direct store access.

6.4 Audit records and AI provenance

Append important changes with enough context to reconstruct who changed what and why.

Required event fields

Event ID and linked request/asset/transfer/AI interaction IDs
Actor account, active profile and department
Timestamp with timezone; display campus times in Asia/Dubai
Action, result and reason
Relevant previous and new values
Evidence/input references and review outcome where applicable

Access and corrections

Workflow administrators review relevant workflow history; system administrators review system audit/recovery records. Ordinary users cannot delete audit history. Corrections append linked entries.

AI review record

Retain input references/snapshot, output, time, provider, model/version or Version unavailable, prompt version, confidence or Unavailable, validation outcome and accept/edit/reject/override record. A score is not a correctness probability.

Synthetic event	Example value
Event Id	AUD-0101
Time	2026-10-30T14:20:00+04:00
Actor	U-WA03 (Workflow administrator)
Record	REQ-0101 / MON-014
Action	Reservation approved
Before	Pending; hold MON-014 active
After	Approved; same hold retained; ownership unchanged
Result	Committed

Reason

Eligible serviceable unit and valid destination; proposer U-RQ07 differs from approver.

Coverage

Record important asset, request, approval, receipt, service, disposition, reporting-input and account changes. Record authorization denials and recovery actions as a proposed diagnostic extension without storing credentials or forbidden payloads.

Drivers: FR12; AI07; BR06; NFR04, NFR10. Never log passwords, AI keys, session secrets or unauthorized payloads.

6.5 Backup and recovery

Required snapshot before every demonstration; isolated restoration target: all snapshot records within 10 minutes.

Snapshot schedule and location

Required: a consistent snapshot before each demonstration (NFR09). Proposed extra: also take one at the end of each day in which records change.
Store timestamped sets under a restricted backups directory outside the web-accessible tree; keep a private off-device copy where available. Only the system-administrator recovery procedure may access them. Never place backups or credentials in the public Git repository.
One coordinated set containing the SQLite snapshot, protected evidence files, audit/AI/notification records, schema/runtime-version information and a file/count manifest. Back up password hashes as protected account data; exclude plaintext secrets and provider keys.
Proposed prototype retention: keep demonstration snapshots and the most recent daily snapshots throughout the course, with access restricted. Live-data retention remains subject to institutional review.

Consistency and restore procedure

Pause writes while the application produces the database snapshot and matching evidence-file set; record one snapshot identifier and verify the manifest before resuming writes. Do not copy independently changing database/evidence files and assume they match.
1. Start a timer and select a complete snapshot/manifest; restore to an isolated database and evidence directory, not over the active demonstration data.
2. Check database integrity, record counts, file manifests and referenced evidence; inspect representative assets, holds, approvals, transfers, audit, AI and notification records.
3. Exercise sign-in and a permitted read, verify a forbidden action is denied, and apply normal expiry checks before stock-sensitive actions.
4. Stop the timer; record snapshot ID, tester, environment, expected/actual counts, elapsed time and evidence. Acceptance requires every snapshot record to be restored within 10 minutes; later changes may be lost.

ADR-12 Backup strategy

Options: full coordinated DB/evidence snapshots OR incremental backups with version tracking and replay.
We choose full snapshots because the small prototype makes one inspectable recovery set practical. We give up smaller incremental backups and changes after the selected snapshot; retained full copies use more storage.
Planned check: restore to an isolated database/evidence directory, verify every record, file reference and access check, and record elapsed time against the 10-minute target.

FR12-FR13; BR02, BR06; NFR09-NFR10, NFR12. Restricted off-device copies are a proposed extra, where available. Backups and secrets never enter the public repository.

7.1 Register an asset

Role: Asset custodian - assigned department | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Register an asset

Asset / unit ID: UNIT-M001	Name / short description: 24-inch HDMI monitor
Category: Electronics	Subcategory: Monitors
Quantity: 1	Condition: Serviceable
Owner department: Mechanical Engineering	Custodian: C-101
Location: Lab M-204	Purchase date: 2024-09-01
Estimated value: AED 300.00	Evidence: front.png - 1.2 MB

Save draft | Request release

Classification suggestion

Suggested fields

Electronics / Monitors; tags: display, HDMI. Material: Unknown.

Confidence

Unavailable - review required.

Reason and source

The title and front.png show a monitor. Unseen ports and power use remain Unknown.

Your decision

Review or edit the fields before saving. A draft is not published.

Review / edit | Ignore

If AI is unavailable

Keep all entered fields. Select the category manually and save the draft; release still needs workflow-administrator approval.

Decision: The custodian reviews classification and saves a draft or proposes release.

Phase 3: FR01-FR02, FR13, FR17; AI02, AI07-AI08; UC02, UC10

Screen 1 of 8

7.2 Submit a resource request

Role: Requester - own request | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Submit a resource request

Purpose: Teaching laboratory	Category / specifications: Monitor / 24-inch HDMI
Quantity / preferred condition: 1 / Serviceable	Urgency: Normal
Needed date: 2026-11-11	Destination department: Chemistry
Receiving custodian: C-202	Destination location: Lab C-107
Selected unit: UNIT-M001	Current requestable quantity: 1 - checked again on confirm

Application rules

Confirm creates Pending and a stock hold. Seeking saves a need without reserving stock.

Confirm request | **Save as Seeking**

Assisted request preview

Proposed need

One 24-inch HDMI monitor for the Chemistry teaching laboratory by 11 November.

Confidence

Unavailable - review required.

Reason and source

The preview uses your purpose, quantity, specifications and destination fields.

Your decision

Review every field. No request or stock hold exists until you confirm.

If AI is unavailable

Keep the request fields and selected unit. Use keyword filters or enter the need manually; normal stock and permission checks still apply.

Decision: The requester confirms the exact request, or saves an unmatched need as Seeking.

Phase 3: FR03-FR06; AI01, AI04, AI08; BR01-BR03; UC03, UC12

7.3 Review recommendations

Role: Requester or department representative - permitted listings | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Review recommendations

Need: 1 serviceable HDMI monitor; Chemistry; due 2026-11-11

All shown units passed permission, publication and safety checks.

Eligible unit	Compatibility	Known factors
○ UNIT-M001	92 / 100	HDMI; cost AED 20.00; no repair
○ UNIT-M003	86 / 100	HDMI; cost AED 25.00; no repair
○ UNIT-M011	72 / 100 - weak	HDMI; repair cost Unknown

Scores describe compatibility, not correctness probability.

Top-two gap: 6 points. No default or emphasized top choice.

Continue with selection | Manual search

Why these matches

Illustrative factor contributions

UNIT-M001: suitability 50 + urgency 10 + transfer 12 + repair 10 + diversion 5 + carbon 5 = 92 points.

Confidence

Unavailable - review required.

Reason and source

UNIT-M001 meets the HDMI need. Its known transfer cost is lower than UNIT-M003.

Uncertainty

A small score gap and missing inputs need review. Unknown facts are not zero.

If AI is unavailable

Show current keyword/filter results in rule order, labelled Manual search results. Preserve the need; do not invent AI scores or explanations.

Decision: The requester chooses explicitly; submission rechecks stock before making a hold.

Phase 3: FR03-FR05; AI01, AI06-AI08; BR01-BR02; UC09

7.4 Approve a transfer request

Role: Workflow administrator - no self-approval | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Approve a transfer request

Request / requester	Reviewer
REQ-014 / R-201	WA-01 - different from requester
Unit / quantity	Reservation
UNIT-M001 / 1	Pending - hold active
Source department / custodian	Destination / custodian
Mechanical / C-101	Chemistry / C-202
Destination location	Safety / publication
Lab C-107	Serviceable / Published
Hold expiry - Asia/Dubai	Decision reason
2026-11-01 14:00	Meets the recorded teaching need

Application rules

Approval retains the hold. Collection evidence and receiving-custodian receipt are required before custody changes.

Approve with reason | **Reject with reason**

Saved match explanation

Existing request evidence

Illustrated match score: 92 / 100 for UNIT-M001. Read-only explanation from the request.

Confidence

Unavailable - review required.

Reason and source

The recorded HDMI requirement and available quantity match the asset record.

Authority

AI cannot approve. Recheck role, requester, stock, safety and expiry before saving.

If AI is unavailable

Decide using the request, asset facts and evidence. New AI output is unnecessary; the server still enforces every approval rule.

Decision: The administrator approves or rejects with a reason; receipt remains a separate human gate.

Phase 3: FR07-FR08, FR12, FR15, FR17; AI08; BR02-BR06; UC05-UC06

Screen 4 of 8

7.5 Track maintenance

Role: Maintenance staff - assigned service job | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Track maintenance

Job / assigned staff	Asset
JOB-008 / MS-011	UNIT-M007 - HDMI monitor
Reported defect	Serviceability
Intermittent display flicker	Under service - allocation blocked
Inspection date	Recorded finding
2026-10-30	Power board fault
Service state	Repair estimate
Inspected	AED 50.00
Evidence	Repair authority
inspection_008.pdf - 0.6 MB	Custodian C-101 - pending decision

Application rules

Saving an inspection does not clear safety. Repair authorization and recorded human clearance remain required.

Save inspection | **Request repair review**

Saved custodian advice

Suggested comparison

Compare repair at AED 50.00 with evidenced replacement at AED 300.00.

Confidence

Unavailable - review required.

Reason and missing inputs

A repair may extend service. Carbon factors and revised retirement date are unavailable.

Authority

Advice is read-only here. Staff record findings; the custodian authorizes repair. AI cannot clear safety.

If AI is unavailable

Continue the inspection and service form manually. Preserve findings and evidence; the asset remains blocked until proper human clearance.

Decision: Assigned staff record the inspection; the custodian reviews repair authorization.

Phase 3: FR09-FR10, FR13-FR14, FR17; AI03, AI08; NFR13; UC07, UC11

7.6 Review sustainability dashboard

Role: Sustainability officer - environmental and life-extension scope | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Review sustainability dashboard

Period: 2026-10-01 to 2026-10-30 | Evidenced completed activity

Stock and open needs **8 available; 3 open needs**	Reuse - M02 **4 units; 4 / 20 = 20%**
Waste diversion - M03 **3 / 4 = 75%**	Estimated carbon - M04 **40 kg CO2e; coverage 2 / 4**
Estimated extension - M05 **120 days / 2 assessed units**	Observed extra service **60 days; separately recorded**

Application rules

Count distinct, evidenced units. Sample carbon factors: 25 - 5 = 20 kg CO2e per covered unit; missing factors stay Unavailable.

Review environmental inputs | **Request summary**

Activity summary draft

Illustrated AI result

Four units were reused. Three of four final outcomes were diverted from disposal.

Confidence

Unavailable - review required.

Reason and source

The draft uses application totals M02-M04. The 40 kg estimate covers only 2 of 4 reused units.

Your decision

Check environmental evidence and coverage. AI cannot create missing factors or count planned actions.

If AI is unavailable

Keep the calculated indicators, source links and coverage visible. Show Summary unavailable; environmental input review still works.

Decision: The sustainability officer validates assigned evidence and reviews the narrative against totals.

Phase 3: FR18; AI05, AI07-AI08; BR07; M02-M05; NFR11; UC13

Screen 6 of 8

7.7 Generate a financial report

Role: Finance officer - financial input and report scope | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Generate a financial report

Reporting period: 2026-10-01 to 2026-10-30

Report type: Avoided purchases and net savings

Replacement cases and source costs - AED

Case	Avoided value	Transfer	Repair / refurb.	Net savings
TR-006	300.00	20.00	30.00 / 0.00	250.00
TR-007	200.00	Unknown	0.00 / 0.00	Unavailable

Known-case subtotal: AED 250.00 | Coverage: 1 of 2 cases

Complete net total: Unavailable - TR-007 has an unknown cost.

Validate financial inputs | Request narrative

Financial narrative draft

Illustrated AI result

Known net savings are AED 250.00 for 1 of 2 cases. The complete total is Unavailable.

Confidence

Unavailable - review required.

Reason and source

TR-006: 300 - 20 - 30 - 0 = 250. TR-007 lacks an evidenced transfer cost.

Your decision

Review figures, period, case links and assumptions. Reject any invented or mismatched amount.

Use reviewed draft

If AI is unavailable

The numeric report remains available. Show Draft unavailable and retain period and criteria; no unchecked narrative is presented.

Decision: The finance officer validates financial inputs and reviews the draft against the numeric report.

Phase 3: FR12, FR18; AI05, AI07-AI08; BR07; M01; NFR11; UC13

7.8 Use the conversational assistant

Role: Requester - authorized catalogue and own requests | Synthetic example; AI output illustrated

Campus Item Exchange Notifications Sign out

Use the conversational assistant

You

Find one HDMI monitor for Chemistry by 11 November.

Assistant

UNIT-M001 fits the recorded need. Source: catalogue UNIT-M001. No approved campus collection policy is available.

Exact request preview - no hold yet

UNIT-M001; quantity 1; purpose Teaching laboratory; 24-inch HDMI; Serviceable; Normal urgency; needed 2026-11-11.

Destination: Chemistry; custodian C-202; Lab C-107.

Confirm request | Edit preview | Cancel

Facts, suggestion and control

Confidence

Unavailable - review required.

Reason and source

The authorized asset record lists HDMI, serviceability and one requestable unit.

Application rules

Recheck permission, current stock, dates and destination when you confirm. The assistant cannot reserve stock.

Your decision

Edit or cancel the preview, or explicitly confirm it through the ordinary request validation.

If AI is unavailable

Retain the question and preview. Show Assistant unavailable and provide the ordinary catalogue, status and request screens.

Decision: The requester confirms the exact preview; only normal validation can create a request and hold.

Phase 3: FR03-FR06, FR16; AI04, AI07-AI08; BR01-BR02; UC12

Screen 8 of 8

AI-use and review statement

Tool use is disclosed separately from the team review and implementation acceptance still required.

Item	Disclosure
Tool	OpenAI Codex.
Drafting	Assisted with architecture choices, AI service policies, approval points, access rules, recovery procedures and decision-record wording, using the supplied Phase 3 requirements and Phase 4 lecture.
Visuals	Generated vector context/DFD and architecture figures, tables and eight synthetic screen wireframes. Reference images guided the blue styling, sparse diagram detail and Level 2 dotted boundary.
Document checks performed	Compared context and Level 1 external flow names/directions/entities and the six Level 2 boundary flows; reviewed six process/five store mappings, all nine profiles plus AI, stated deadlines and example arithmetic; rendered and visually checked pages. These are checks of the design document.
Human responsibility	Each member must verify the design and requirement interpretation, review AI-assisted content, and commit their own contribution. Provider access, capabilities, institutional rules and all planned implementation/acceptance checks still require actual evidence.

Source documents

Campus Item Exchange Phase Three Requirements Document, 4 October 2026 (Phase_3_04_Cumulative.docx).
Phase 4 Architecture Design, Fall 2026 (Phase_4_Architecture_Design_fall26_Blackboard.pptx), especially slides 3-4 and 15-24.

AI suggestions never replace the nine-profile permission matrix, required human approval, recorded safety clearance or evidence-based reporting.

Phase 5 Detailed Design

Campus Item Exchange | Data, behavior and implementation contracts | 30 October 2026

Purpose and scope

This section develops the Phase 4 architecture into data records, classes, interactions, state rules and interfaces. It retains one Flask application, SQLite, protected evidence files, nine permission profiles and advisory AI. The designs describe intended behavior; implementation and measured acceptance results belong in later phases.

Section	Pages
8 Data model	30-31
9 Class model	32
10 Sequence models	33-34
11 Activity and AI workflows	35-36
12 State models	37-39
13 Components and deployment	40-41
14 Use cases and requirement links	42
15 Interfaces and implementation checks	43-45

Design rules

A listing groups physical units. Each hold and transfer identifies the actual unit; approval does not change its owner or custodian.

People retain authority

The nine-profile access matrix and human gates in Sections 5 and 6 apply to every class, workflow and interface. A system-administrator profile alone grants no business approval.

AI remains advisory

Section 4 controls all five AI services. Confidence, compatibility and review gates remain separate; ordinary application rules recheck every confirmed action.

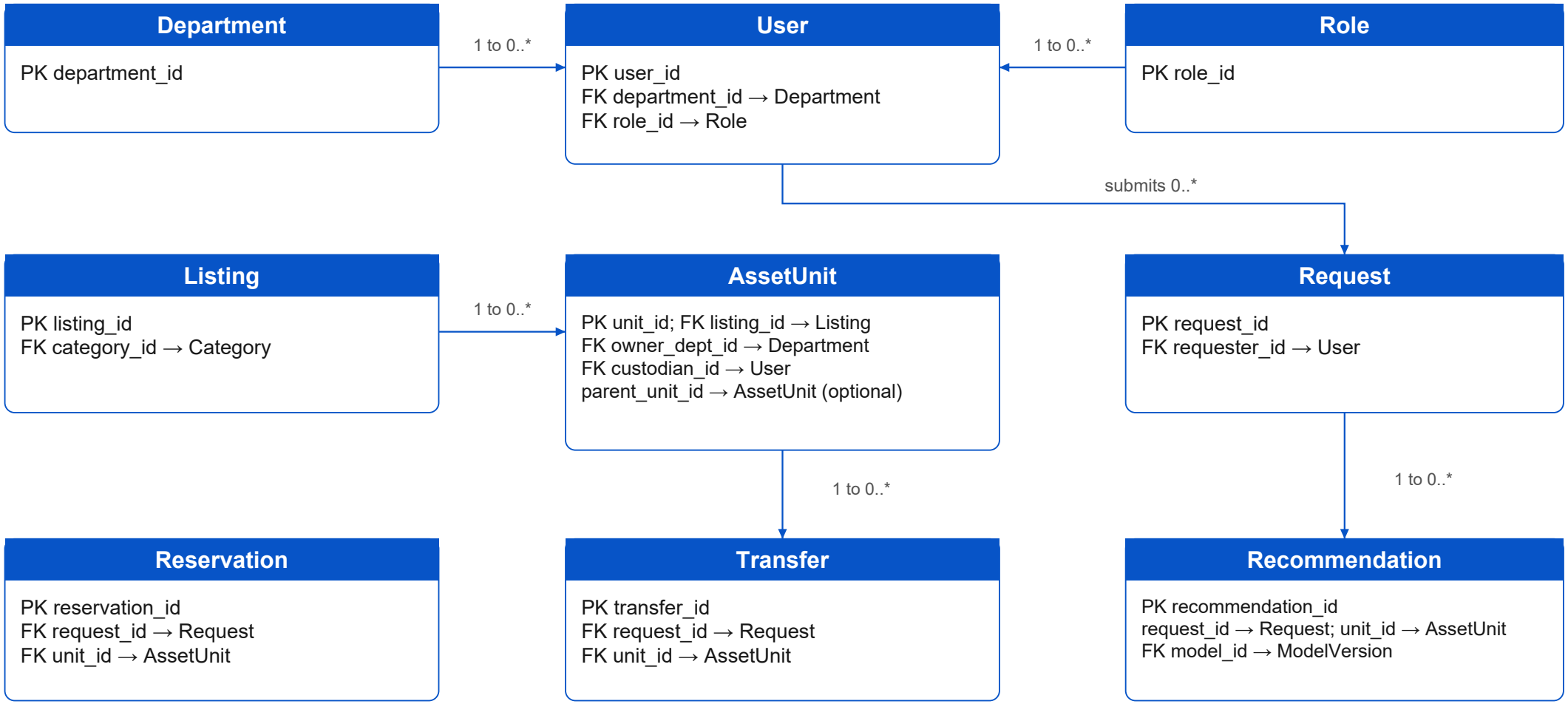
Evidence and review

Examples remain synthetic. Model diagrams and planned checks are design specifications. Team review, individual contributions and implementation results still require actual evidence.

Baseline: Phase 4 architecture, Sections 1-7; Phase 3 requirement references retained throughout.

8.1 Entity relationship model

A listing groups physical units; reservations and transfers identify each unit individually.



PK = unique row. Required FKs link one child to one parent; each parent can have 0..* children. Each Reservation and Transfer links one Request and one AssetUnit. Matching recommendations require both; other AI tasks may omit them. Referenced history uses RESTRICT deletion.

Drivers: FR01-FR14, FR17-FR18; AI07; BR01-BR07; NFR05, NFR10-NFR11, NFR13.

8.2 Supporting records and data constraints

Keys preserve evidence, review decisions and the history of each physical unit.

Record and primary key	Foreign keys and principal fields
Category	category_id; optional parent_id → Category; name. Listing selects one category.
Approval	approval_id; exactly one request_id, listing_id or disposition_id → its parent; actor_id → User; gate, decision, reason, time.
Inspection	inspection_id; unit_id → AssetUnit; inspector_id → User; outcome, evidence and date.
Maintenance	maintenance_id; unit_id → AssetUnit; assignee_id → User; status, action, cost and condition.
Disposition	disposition_id; unit_id → AssetUnit; proposed action and status.
LifeEvent	event_id; unit_id → AssetUnit; responsible_id → User; event type and outcome.
Impact	impact_id; unit_id → AssetUnit; event_id → LifeEvent; costs/factor sources and prior/revised retirement dates.
Audit	audit_id; actor_id → User; target references, before/after values, reason and time.
Evidence	evidence_id; uploader_id → User; protected path and parent-record link.
Notification	notification_id; recipient_id → User; event_id → Audit; optional read_at.
ModelVersion	model_id; provider, model identifier, optional provider version and prompt version.

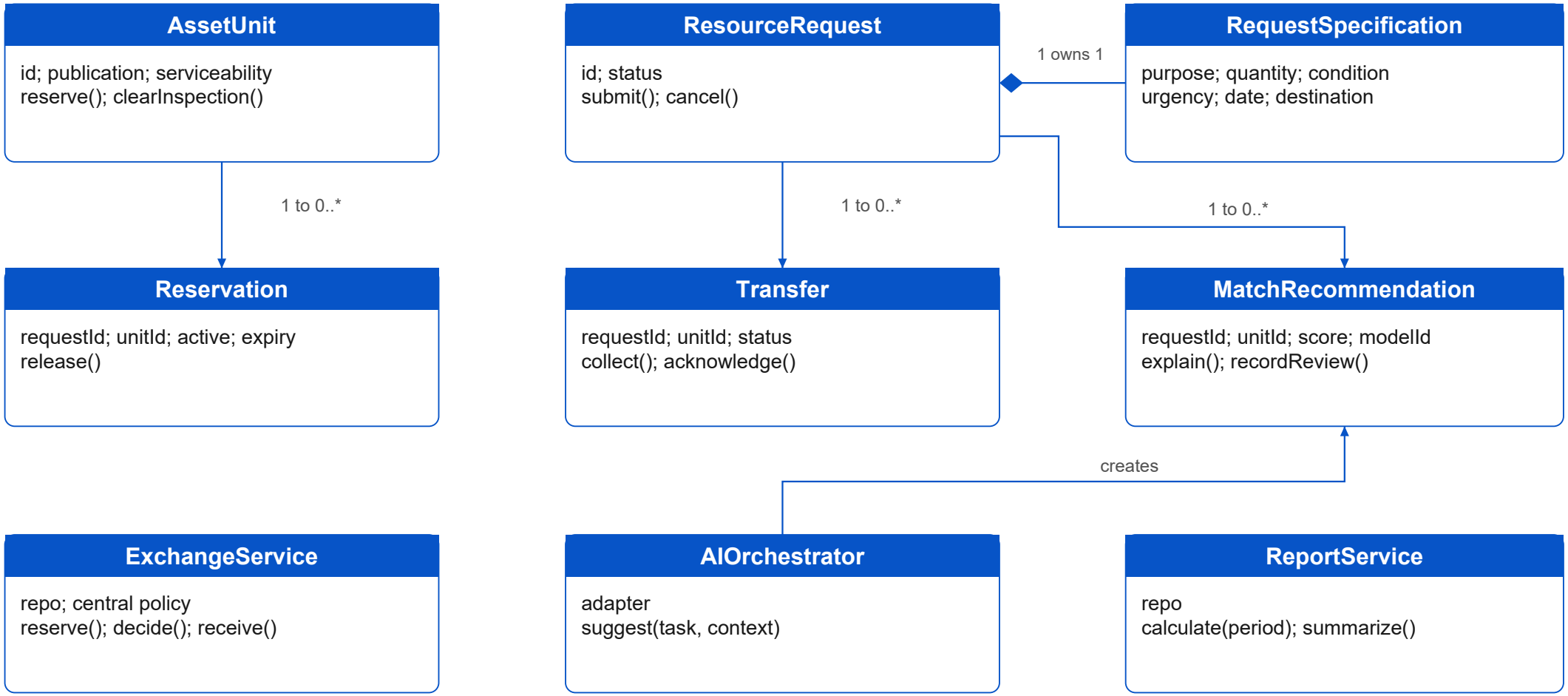
Allocation and history constraints

Enforce one active hold per unit with a conditional unique index. Transfer has UNIQUE(request_id, unit_id). Multi-unit requests reserve and complete all units atomically. Actor fields reference User; owner fields reference Department. Recovered components receive new IDs linked to their parent. Corrections append history.

Drivers: FR09-FR18; AI07-AI08; BR06-BR07; NFR04-NFR05, NFR10-NFR11, NFR13.

9 Class model

Services enforce permissions and transactions; domain objects reject invalid transitions.

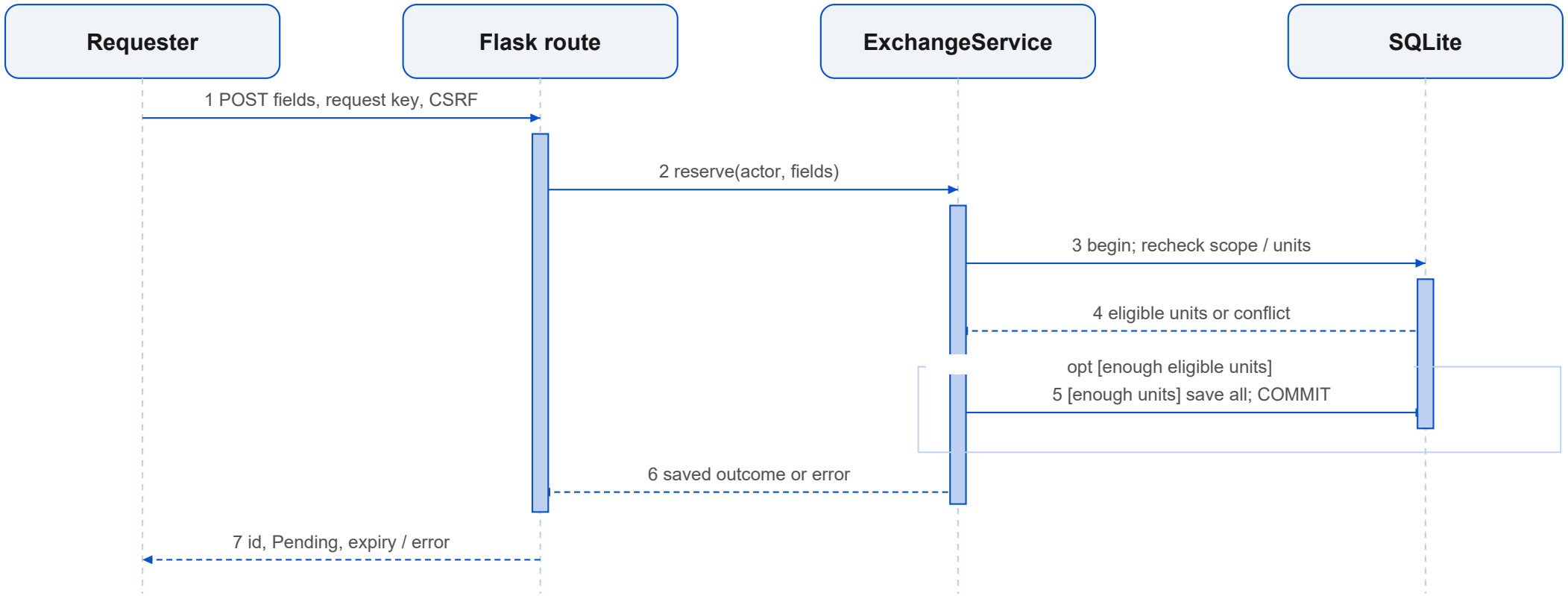


User submits 0..* requests and has custody of 0..* units. Each Reservation, Transfer and MatchRecommendation references one request and one unit. MaintenanceJob has status, optional cost, inspect() and close(); one unit has 0..* jobs. Composition does not permit deletion of retained history. See Section 12.3 for operation guards.

Drivers: FR04-FR14, FR17-FR18; AI06-AI08; BR01-BR07; NFR05, NFR11, NFR13.

10.1 Sequence model for request reservation

Explicit confirmation creates the request and all required stock holds in one transaction.



Atomic outcome and alternatives

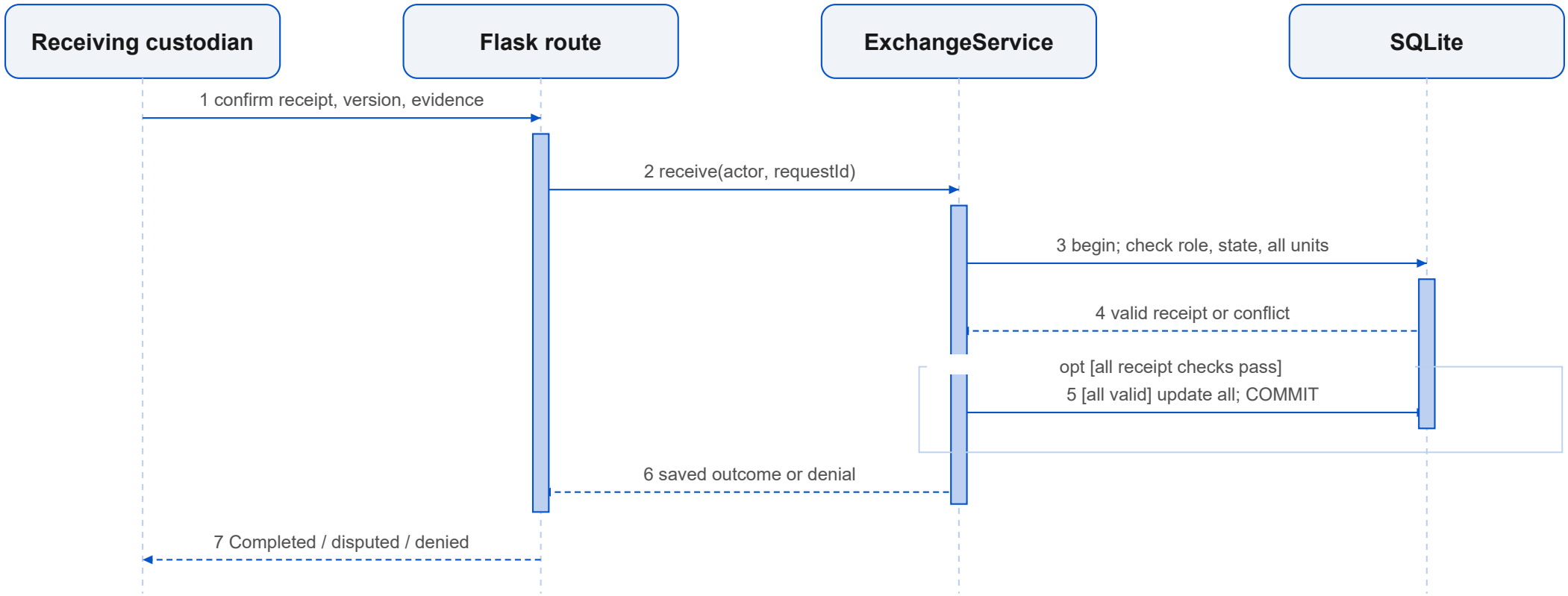
Success saves the Pending request, all holds, audit and notification events together. If any unit is unavailable, roll back and return 409; retain valid form input. Invalid input returns 400 and forbidden actions return 403.

The same request key returns the saved result. Saving an unmatched need as Seeking creates no hold or expiry. No AI network call runs while a write transaction is open. Solid arrows are calls; dashed arrows are returns.

Drivers: FR03-FR06; BR01-BR03; NFR05; UC03-UC04.

10.2 Sequence model for receiving units

Ownership and custody change only after the complete approved receipt is confirmed.



One complete handover transaction

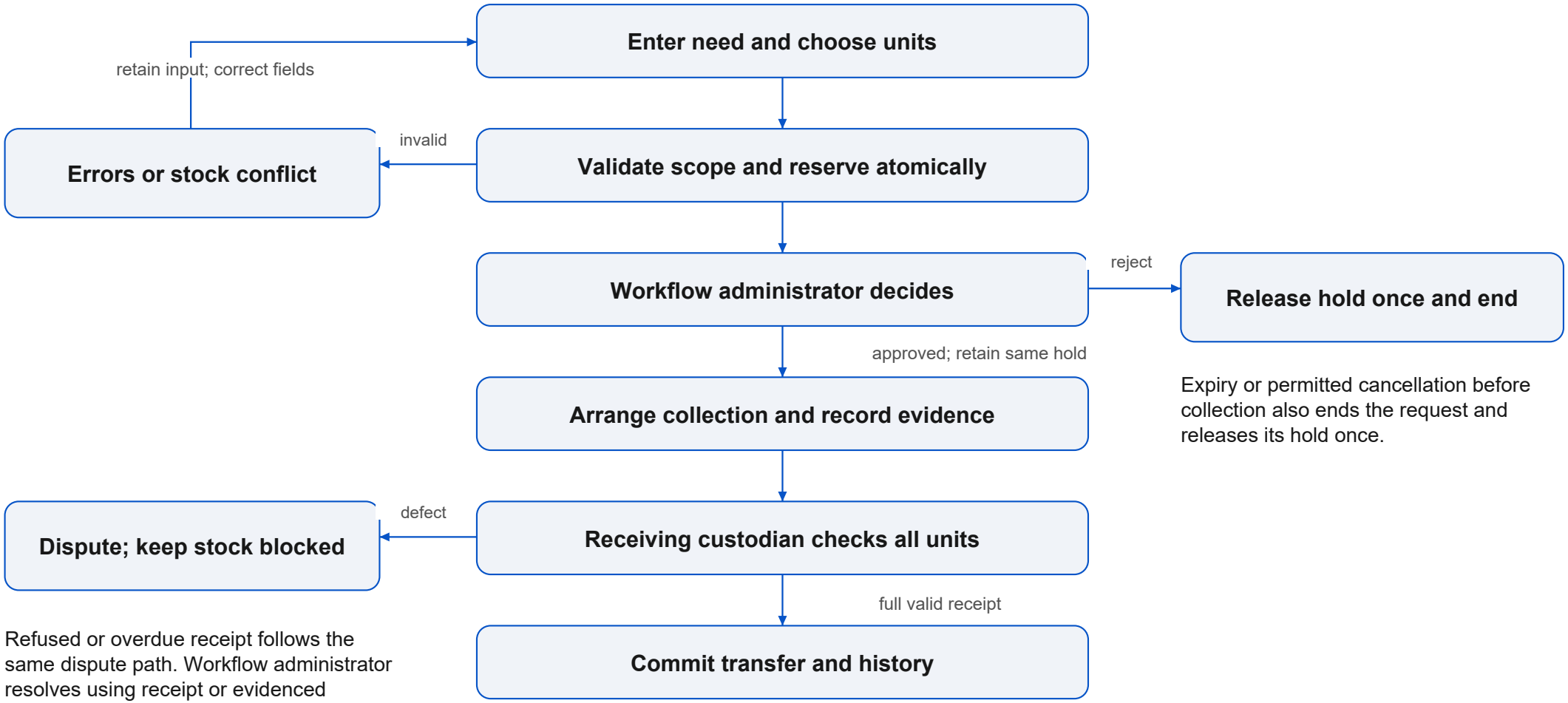
Complete the request and unit transfers; update owner, custodian and location; close holds; append audit, life and notification events. A failure before COMMIT rolls back every update. Repeating a completed receipt returns the existing result.

Refused, defective or overdue receipt enters dispute and keeps units blocked. Source collection evidence is a precondition. A workflow administrator resolves the dispute through evidenced receipt or physical return; AI cannot confirm a physical handover.

Drivers: FR07-FR08, FR12-FR13, FR17; BR02-BR06; NFR05; UC05-UC06.

11.1 Exchange activity workflow

Control flow separates request approval, physical collection and confirmed receipt.



Expiry or permitted cancellation before collection also ends the request and releases its hold once.

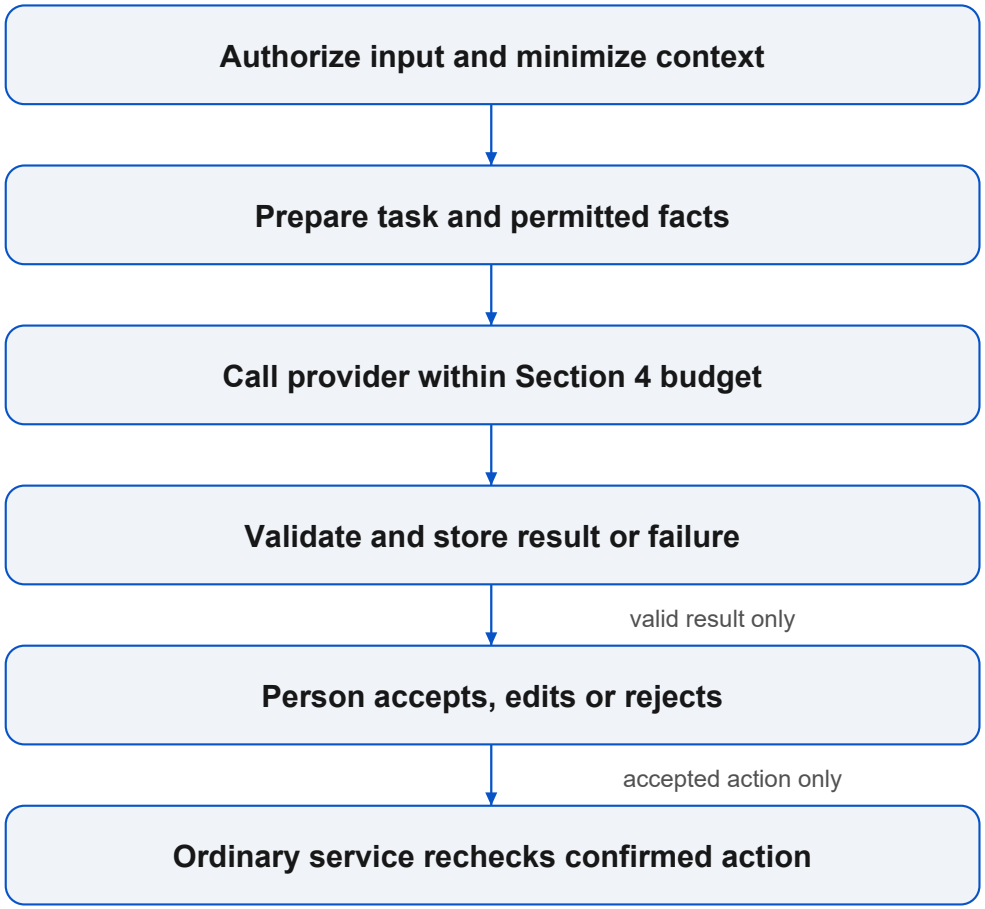
Refused or overdue receipt follows the same dispute path. Workflow administrator resolves using receipt or evidenced physical return.

Approval changes neither ownership nor custody. A dispute never completes or releases stock automatically. Vertical arrows show control flow; they are different from the data flows in Section 2.

Drivers: FR03-FR08, FR12-FR13, FR17; BR01-BR06; NFR05, NFR13.

11.2 AI recommendation workflow

Validate supplied facts and retain ordinary human and server checks for every confirmed action.



Task-specific checks

For matching and ranking, exclude unauthorized, unsafe, unavailable or incompatible units before scoring. Classification and advice use their own authorized asset evidence. Validate schema, IDs, facts, source revisions and report totals.

Failure and uncertainty

Invalid or timed-out output becomes a labelled manual fallback. Unknown facts remain Unknown. Confidence below 0.80 or Unavailable needs review. Compatibility below 75/100 is weak. Top-pick emphasis needs complete required facts and a 10-point gap when two candidates exist.

Deadline and provenance

Matching uses at most 6 seconds; ranking uses at most min(6 seconds, time left), within one shared 12-second deadline. Keep all other Section 4 budgets. Save input references, provider/model and prompt versions, output, validation and review outcome.

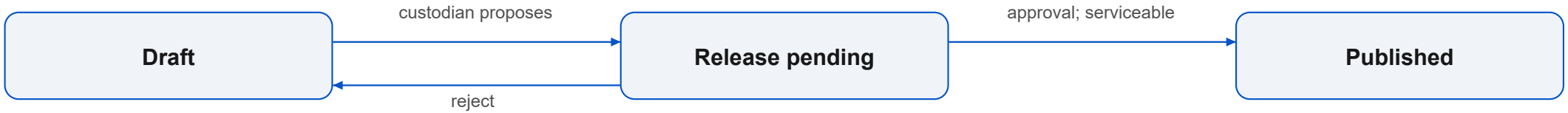
Acceptance gives no extra permission. A valid suggestion may fail a later state or stock check. No automatic retry or late result may change records; reports remain numeric and usable when optional AI narrative is unavailable.

Drivers: AI01-AI08; FR05, FR12, FR16-FR18; NFR03-NFR06, NFR10, NFR14; UC09-UC13.

12.1 Asset state models

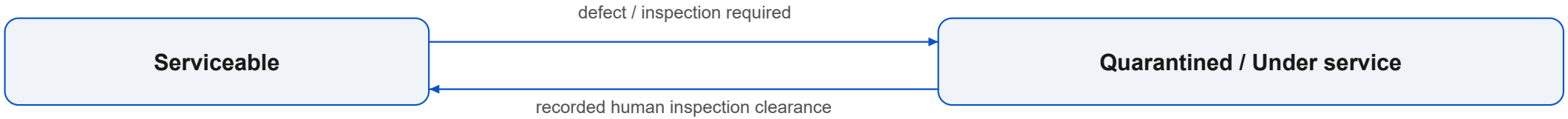
Publication, serviceability and lifecycle are independent facts; Published never overrides unsafe condition.

Publication

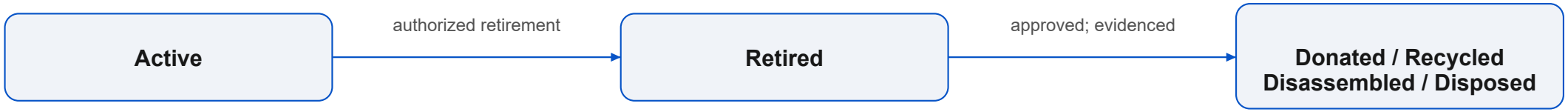


Withdrawal returns Published to Draft; republishing requires a new approval. Workflow administrator decisions require a reason and no self-approval.

Serviceability



Lifecycle and final disposition



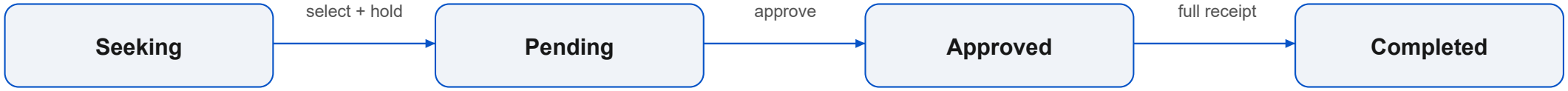
Retirement blocks allocation. Held or allocated units cannot enter disposition. Final outcomes are terminal; recovered components receive new linked unit IDs. AI cannot approve retirement, execute disposal or clear safety.

Drivers: FR01-FR02, FR09-FR10, FR13-FR14, FR17; AI03, AI08; BR01, BR04-BR07; NFR13.

12.2 Request and transfer state models

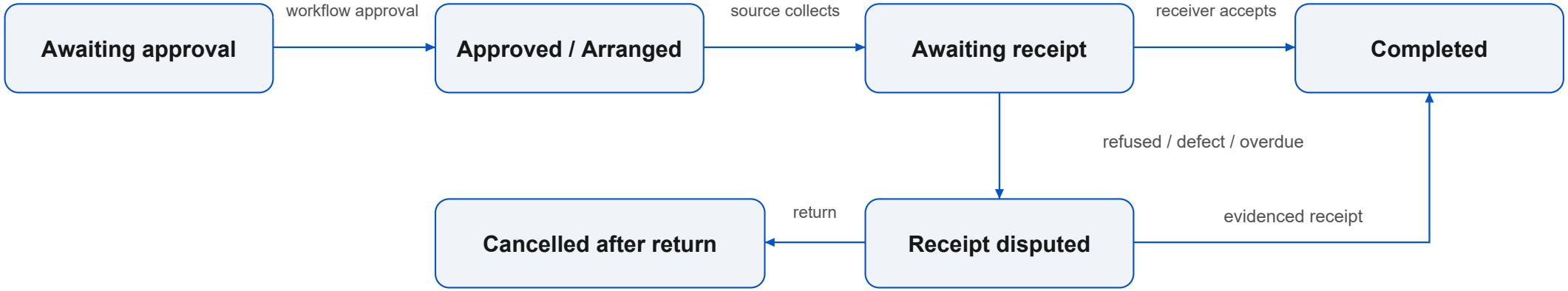
Holds remain active through approval and unresolved collection or receipt disputes.

Request status



Seeking has no hold or expiry and may cancel. Pending may reject, expire or cancel. Approved may cancel or expire only before collection. Approved retains the same hold; full valid receipt completes. Rejected, Expired, Cancelled and Completed are terminal and cannot reopen.

Transfer status

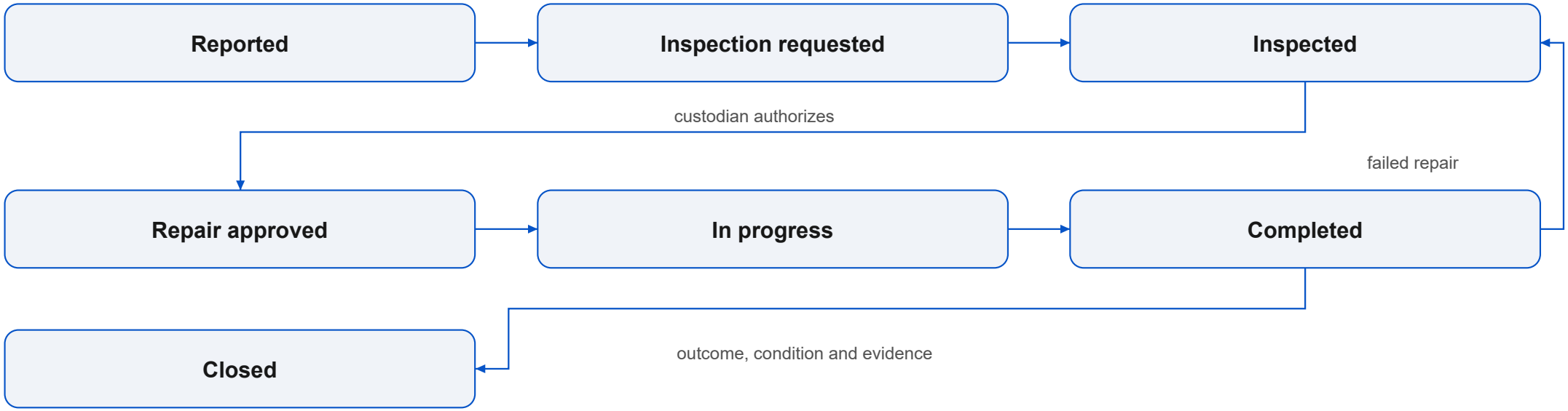


Before collection, lawful cancellation releases holds once. After collection, a workflow administrator must resolve the dispute through evidenced receipt or physical return. An unresolved dispute never frees stock. Every decision preserves actor, time and reason.

Drivers: FR04-FR08, FR12-FR13, FR17; BR02-BR06; NFR05; UC03-UC06.

12.3 Maintenance states and operation guards

Custodian repair authorization and recorded inspection evidence control service completion.

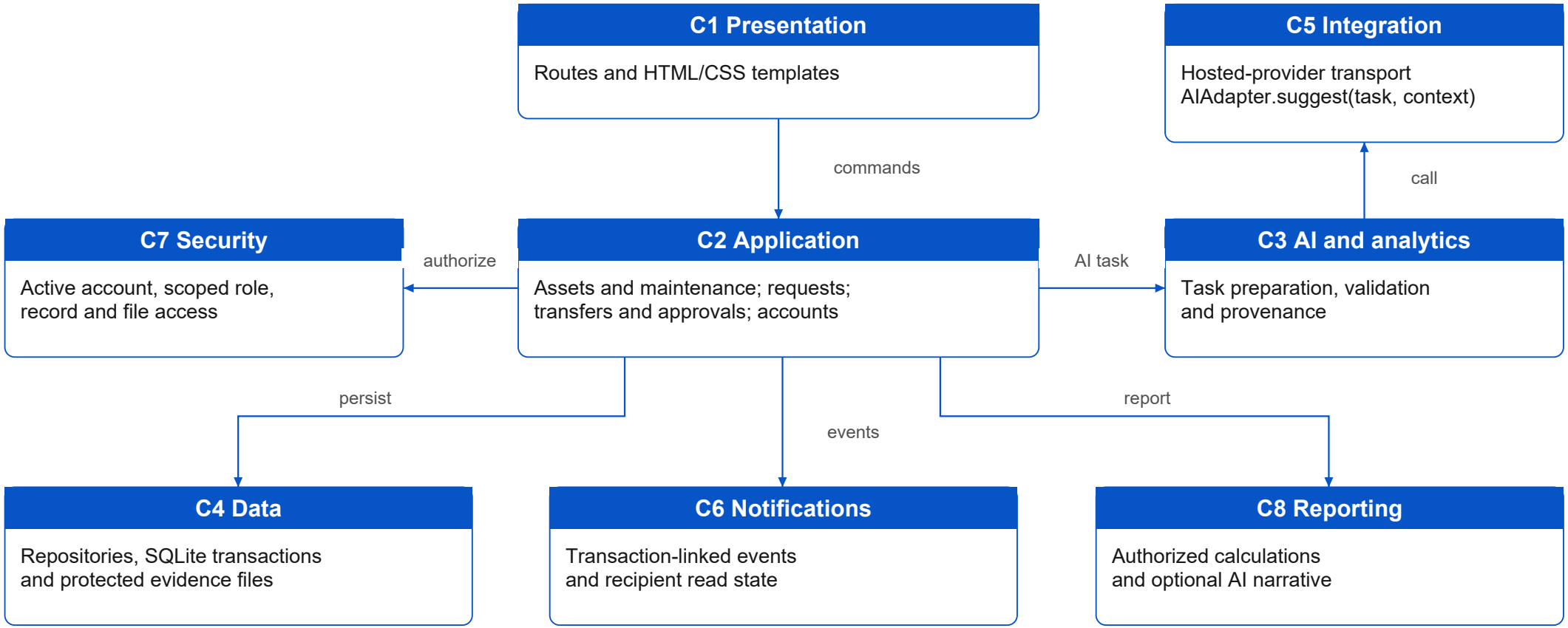


Operation	Required precondition and invariant
ExchangeService.reserve	Authorized creator; positive quantity; every requested unit eligible. Request and holds commit together or all fail.
ExchangeService.receive	Receiving custodian; full approved receipt. Ownership, custody, location, transfers, holds and history commit together.
MaintenanceJob.close	Assigned service user; authorized repair; completion evidence and updated condition. Closure alone cannot clear an unsafe unit.
ReportService.calculate	Authorized completed events; deduplicate units and costs. Missing inputs remain Unavailable.

Drivers: FR04-FR14, FR17-FR18; AI03; BR01-BR07; NFR05, NFR11, NFR13; UC07, UC11.

13.1 Components and interfaces

Dependencies point from consumer to provider. One Flask deployment retains the eight Phase 4 components.



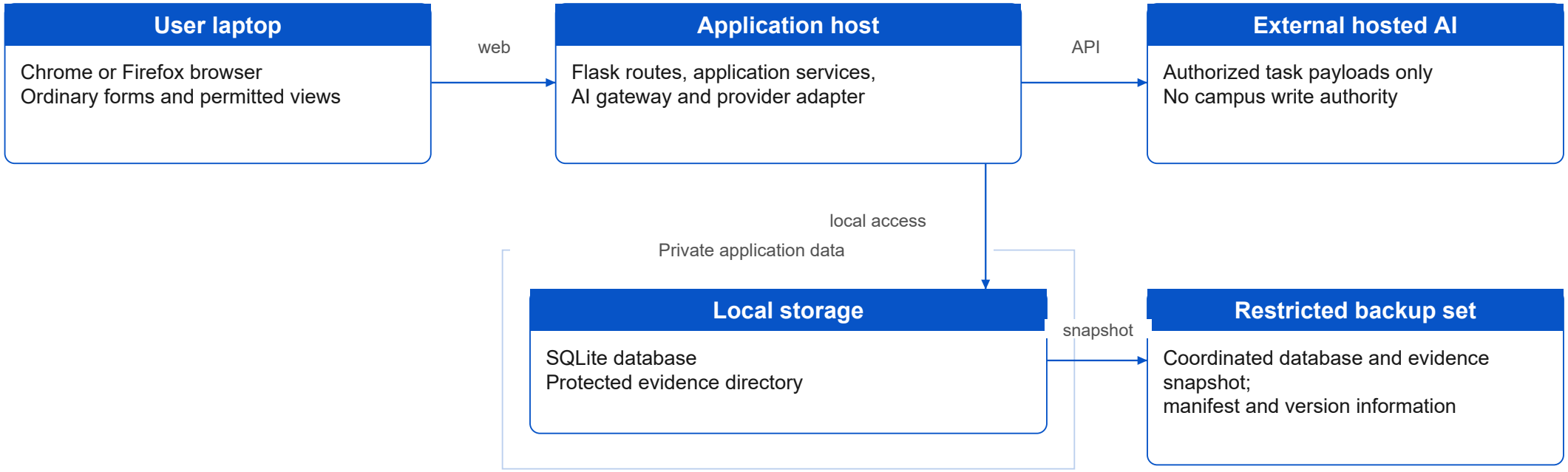
Boundary rules

Routes call application services; they never write tables directly. C7 and scoped repositories protect every server path. C8 reads through C4 and requests optional narrative through C3. C3 calls the provider only through C5, outside write transactions. The provider cannot change records or report arithmetic.

Drivers: FR01-FR18; AI01-AI08; BR01-BR07; NFR03-NFR06, NFR10-NFR12.

13.2 Deployment view

The prototype keeps the application, SQLite and protected evidence together; hosted AI remains external.



Connection and storage rules

Keep SQLite and evidence on the application host, rather than a shared writable network database. Use TLS for a remote demonstration; localhost HTTP is limited to local development. Secrets remain in host environment settings. Record the actual operating system, browser and runtime versions during validation.

Recovery

Take a consistent snapshot before every demonstration. Restore all snapshot records and evidence to an isolated location within 10 minutes. Daily snapshots and a private off-device copy, where available, remain proposed extras. Later changes may be lost.

Drivers: FR12-FR13; NFR02-NFR04, NFR07-NFR10, NFR12, NFR15. Recovery follows Section 6.5.

14 Use cases and requirement links

The use case IDs are retained from the drafts. Grouped roles never merge the permissions in Section 6.

ID	Use case	Role and control
UC01	Sign in and sign out	Active campus accounts.
UC02	Register and publish asset	Custodian prepares; workflow administrator decides publication.
UC03	Search and request asset	Requester / department representative, within scope.
UC04	Track and cancel request	Permitted request access; creator and state restrictions apply.
UC05	Decide request	Workflow administrator; no self-approval.
UC06	Confirm handover	Source and receiving custodians; administrator resolves disputes.
UC07	Maintain and dispose asset	Custodian, assigned maintenance staff and workflow administrator at separate gates.
UC08	Manage accounts and departments	System administrator.
UC09	Find AI matches	Requester / department representative; advisory AI provider.
UC10	Review AI classification	Custodian; advisory AI provider.
UC11	Review sustainability advice	Custodian; advisory AI provider.
UC12	Use natural language assistant	Permitted users within ordinary record scope; advisory AI provider.
UC13	Generate activity report	Authorized reporting officers; workflow administrator for operations; optional AI draft.

Requirement links: Data and controls support FR01-FR18 and BR01-BR07; classes and state/sequence models develop UC01-UC08; AI workflows and contracts develop UC09-UC13 and AI01-AI08. Deployment and planned checks address the NFR baseline. The AI provider only assists UC09-UC13 and cannot authorize campus actions.

Drivers: UC01-UC13; FR01-FR18; AI01-AI08; BR01-BR07; NFR01-NFR15. Each detailed view lists its principal drivers.

15.1 Interface contracts

All commands pass the ordinary identity, scope, state and evidence checks before changing records.

Interface	Input and output contract
POST /requests	Authenticated actor, idempotency key, need fields, selected listing/units and quantity. Return 201 with id, status and expiry; 400 invalid input; 403 forbidden; 409 stock conflict. Seeking creates no held units or expiry.
POST /requests/{id}/decision	Workflow administrator, expected state/version, approve or reject, and reason. Return the saved decision; 409 for stale state. Deny self-approval.
POST /transfers/{id}/receipt	Receiving custodian, expected version, full acknowledgment and evidence ID. Complete the request only when every approved unit receipt validates. A retry returns the saved outcome.
AIAdapter.suggest(task, context)	Task enum, authorized facts, schema, prompt version and deadline. Return structured output, model/version, confidence or null, evidence IDs, latency and usage. Invalid output becomes a typed failure; budgets remain those in Section 4.
GET /reports?from=&to=	Scoped actor and inclusive reporting period. Return calculated totals, input coverage, source IDs and labelled assumptions. An optional AI narrative cannot change the arithmetic.

Failure and repeat behavior

Use protected sessions, CSRF tokens, scoped authorization and validated fields. Reject stale versions or invalid states without partial writes. Repeating a successful confirmation returns the existing result. Valid form input remains available after an error.

Drivers: FR03-FR08, FR12-FR13, FR16-FR18; AI01-AI08; BR01-BR07; NFR03-NFR06, NFR10-NFR11.

15.2 Scoring and reporting rules

Eligibility is checked before scoring. Calculations use evidenced completed events and remain independent of AI.

Eligibility first

Exclude unauthorized candidates, incompatible required specifications, insufficient quantity, unavailable dates and unsafe units. A high score cannot override a failed rule.

Known factors and coverage

Suitability considers semantic relevance, category, purpose, condition and specifications. Explain urgency, transfer affordability, repair readiness, diversion and carbon benefit only when evidence exists. Use a weighted mean of available normalized factors; renormalize known weights and show coverage. Unknown is not zero.

Review and explanation

Compatibility below 75/100 is weak. Confidence below 0.80 or Unavailable requires review. Top-pick emphasis also needs a 10-point gap when two candidates exist, complete required facts and no confidence warning. Retain contributions, sources, missing fields and the scoring version. Stable unit ID breaks a display tie; the user still chooses.

Indicator	Calculation
M01 Net savings	Avoided purchase - transfer - repair - refurbishment costs.
M02 Reuse	Unique reused cohort units / eligible cohort units.
M03 Diversion	Unique donated, recovered or recycled units / all final-disposition units.
M04 Carbon	Replacement-scenario emissions - reuse-scenario emissions.
M05 Life extension	Revised retirement date - prior retirement date. Observed extra service stays separate.

Calculation integrity

Prevent double counting of transfers, costs and parent/component units. Zero denominator: Not applicable. Missing input: Unavailable. Keep negative results, reporting period, evidence and coverage visible. AI can draft from these values but cannot invent or change them.

Drivers: FR12, FR18; AI01, AI05-AI08; BR07; M01-M05; NFR11, NFR14. Thresholds remain proposed review gates.

15.3 Planned implementation checks

Run and record these checks during Phases 6 and 7. This design document does not claim implementation success.

Check	Expected evidence
Final-unit reservation race	One valid request obtains the last unit; the competing request returns conflict with no partial holds.
Duplicate confirmations	Repeated request or receipt returns the saved outcome without duplicate stock changes or transfers.
Receipt dispute	Units remain blocked until evidenced receipt or physical return resolves the case.
All nine profiles	Allowed actions succeed; forbidden direct actions, cross-department records and protected files are denied.
Audit and notifications	Actor, reason, evidence links and recipient scope remain consistent with the committed transaction.
AI success and failure	Exercise all five services, timeout, false IDs/facts, wrong totals, stale results and fallback; enforce Section 4 gates.
Report calculations	Verify sources, distinct-unit counting, missing inputs, zero denominators and negative results using known fixtures.
Setup and isolated restore	Record runtime/browser versions; restore snapshot records and evidence within 10 minutes without replacing active data.

For every check, record tester, environment, date, expected result, actual result and evidence. Validate provider/model choice, capacity, evidence handling and institutional policy before acceptance.

AI assistance and source disclosure

OpenAI Codex assisted with consolidating the detailed design, adapting diagrams and checking document consistency. Phase 5 was developed from the four supplied cumulative Phases 4 and 5 drafts and the completed Phase 4 document. The team remains responsible for review, individual contributions and actual implementation testing.

Drivers: FR01-FR18; AI01-AI08; BR01-BR07; NFR01-NFR15. Acceptance requires actual expected/observed results and evidence.